GRADE CHANGE INTELLIGENCE IN PAPER MAKING PROCESS

GradeIQ: Explainable Automatic Grade Change Intelligence & Setpoint Optimizer

Theme: Smart Automation / Industrial Process Intelligence | Honeywell QCS Hackathon Challenge

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Problem Statement ID: Honeywell QCS Challenge Proposed Solution: Ensemble ML + SHAP XAI Advisor

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PROPOSED SOLUTION: GRADEIQ AI INTELLIGENCE LAYER
An intelligent external advisor that enhances Honeywell QCS Machine Direction (MD) Multivariable Model Predictive Control without disrupting plant hardware loops.

• Proactive Off-Spec Risk Prediction (Ensemble ML): Forecasts when the primary quality variable—Basis Weight—is at risk of deviating >2.5% from
setpoint up to 15 minutes prior to violation during auto grade/recipe transitions.

• Source-Tagged Setpoint Optimization: Recommends corrective setpoint adjustments for Stock Flow, Filler Flow, Dryer Steam Pressure, Wire
Vacuum, and Machine Speed to dramatically reduce stabilization time and eliminate broke cull.

• Explicit Inference Provenance: Every recommendation on the operator display is explicitly tagged with its root inference source (e.g., [Historical
Mining: Transition A32->B18], [Recipe Process Limits], [AI New Correlation Discovery]).

• Explainable AI (XAI) Guidance: Integrates SHAP (SHapley Additive exPlanations) to bridge operator skill shortages, showing newer operators
exactly *why* an action is recommended and which sensor parameters are causing instability.

• Closed-Loop Operator Trust Engine: Features a Human-in-the-Loop dashboard with dedicated Accept/Reject controls. All responses are audited
and logged to evaluate recommendation accuracy and adaptively recalibrate the AI models.

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TECHNICAL APPROACH & MODULAR SYSTEM ARCHITECTURE

Core Tech Stack & Methodology
• Machine Learning: Scikit-Learn, XGBoost, Gradient Boosting, &
Random Forest voting ensemble.
• Explainability (XAI): SHAP TreeExplainer real-time decomposition
& conditional covariance matrix mining.
• Dashboard & Deployment: Gradio interactive web UI, Plotly
dynamics, suitable for AWS S3/RDS & EC2 Edge inference.
• Data Engineering: Automated trajectory feature gradients over
rolling 2.5-minute transition windows.

Data Flow Communication: Ingests historical DCS trends, QCS scanner logs, MIS reports, and operator alarms -> Executes Kalman sliding-window feature imputation -> Predicts deviation trajectories -> Surfaces source-tagged explainable guidance.

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FEASIBILITY, INDUSTRIAL VIABILITY & EDGE CASE MITIGATION
Why Feasible & Deployable Today

• Zero Hardware Modification: Sits alongside (not inside) existing
Honeywell QCS MD Multivariable control loops as an external
supervisory advisory layer.

• Reuses Existing Data: Fully leverages existing DCS historians,
scanner diagnostic tables, and recipe limit logs already archived by
mills.

• High Return on Investment: Reducing broke cull material by just 2 minutes per grade change saves mills up to $350,000 annually in energy and wasted chemical fiber.

Manufacturing Edge Cases & Mitigation

• Sensor Latency & Packet Loss: Implements real-time forward-fill
linear imputation and sliding-window Kalman filtering for momentary DCS scanner drops.

• Noisy & Outlier Scanner Data: Robust ensemble voting algorithms prevent transient electrical anomalies from triggering false alarm
deviation alerts.

• Operator Trust & Skill Shortage: SHAP waterfalls translate numerical
logits into plain English explanations, earning junior operator confidence.

• Sparse Rare Transitions: Utilizes synthetic minority time-series data
augmentation (SMOTE-TS) to guide extreme GSM recipe jumps.

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ARTIFACTS & REAL-TIME OPERATOR ADVISORY STUDIO

Deliverables Repository Structure: Includes realistic industrial dataset (paper_mill_grade_change_data.csv), ensemble XAI pipeline, interactive
Gradio/Plotly dashboard studio, and automated Google Colab 1-click execution notebook.

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RESEARCH, REFERENCES & INDUSTRIAL BENCHMARKS
[Ref 1] Honeywell Experion MX & QCS Automatic Grade Change Application Notes: Product documentation on Machine Direction (MD) Multivariable Model Predictive Control, target calculation, and coordinated ramping of paper machine setpoints.

[Ref 2] Lundberg, S. M., & Lee, S.-I. (2017): 'A Unified Approach to Interpreting Model Predictions' (SHAP). Advances in Neural Information Processing Systems (NeurIPS 2017). Foundation for GradeIQ real-time operator root-cause explainability.

[Ref 3] Chen, T., & Guestrin, C. (2016): 'XGBoost: A Scalable Tree Boosting System'. ACM SIGKDD International Conference on Knowledge Discovery and Data
Mining. Used for robust multi-loop trajectory risk classification.

[Ref 4] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016): 'Why Should I Trust You?: Explaining the Predictions of Any Classifier' (LIME). ACM SIGKDD 2016.
Benchmark reference for trust modeling in human-in-the-loop cyber-physical systems.

[Ref 5] TAPPI Journal Case Studies on QCS Control Loop Optimization: Published industrial engineering benchmarks demonstrating that proactive control loop decoupling and faster grade transition ramps significantly reduce broke fiber cull and thermal stabilization losses in commercial paper mills.

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